Evaluating Effectiveness of ADU Policy Approaches Through a Geospatial Analysis of the Montreal Metropolitan Area

Proposal for Honors Research in Geography B.A.

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Introduction

Affordable housing has grown rapidly as a public policy issue in recent years, prompting policy makers to turn to new and more flexible methods of increasing housing stock. One such method is referred to as an ADU, or accessory dwelling unit. ADU is an umbrella term for backyard cottages, garage conversions, in-law suites and the like, usually constructed as a second smaller housing unit on land zoned for single family development. They have increasingly been seen as a fast and affordable way to bring soft densification at the local level (Lessard 2018), and have been spearheaded by states such as California and Oregon in the US as well as British Columbia and Ontario in Canada. These states and provinces have taken varying approaches to their ADU policies, with some focusing on minimum ADU sizes, some on square footage of the lot, and others still on floor area ratios (FAR). Additionally there are many smaller factors that differentiate policies, such as allowing ADUs in front yards and parking requirements (Lessard 2018).

These policies are often created based on political limitations and opinions about the intended form and character of neighborhoods rather than being made to achieve certain goals in housing development. While one study in Southern California compared California legislation with the researchers recommended changes (Kim D. et al. 2023a), there has been no holistic quantitative comparison of ADU policy approaches that could be found. This study attempts to fill this gap by providing an analysis of the advantages and disadvantages of various policy approaches, specifically evaluating how many ADUs could be permitted according to each policy approach as well as the characteristics of the locations in which they could and could not be permitted. This will be done through a geospatial analysis of the Montreal metropolitan area, chosen due to its wide variety of housing stock and lack of current ADU policies in most municipalities. The purpose of this study is to provide policy makers and advocates with quantitative data to inform them as to which policies will be best for their objectives.

Research Aim and Questions

Research Aim

The aim of this study is to evaluate the effectiveness as well as the advantages and disadvantages of various ADU policy approaches. The objective of the study will be to assess maximum potential development, represented through the number of permittable units. To this extent, the study will focus on physical factors such as size restraints, density, and surrounding features such as transit connections. It will not encompass non-physical determinants such as financial ability or public sentiment.

Research Questions

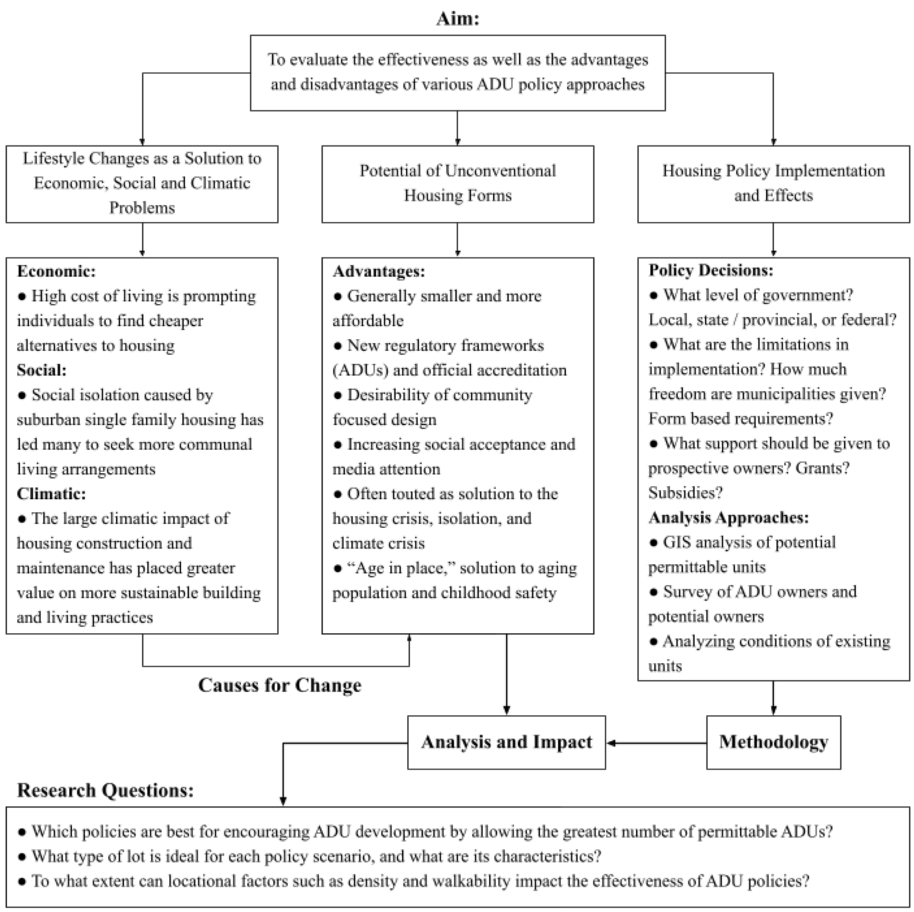
Within the central aim, the study will seek to answer three main questions:

1. Which policies are best for encouraging ADU development by allowing the greatest number of permittable ADUs?
2. What type of lot is ideal for each policy scenario, and what are its characteristics?
3. To what extent can locational factors such as density and walkability impact the effectiveness of ADU policies?

Conceptual Framework

Introduction

This section outlines the conceptual framework used to guide the research. The framework contains three bodies of literature: lifestyle changes as a solution to economic, social, and climatic problems, potential of unconventional housing forms, and housing policy implementation and effects. These bodies of literature shape the context surrounding ADU policy as well as the various forms and implementations of these policies. They serve to guide the direction and impact of the analysis, explaining the importance of research in this domain as well as the broader impacts of research results. The conceptual framework diagram is depicted below:



Lifestyle Changes as a Solution to Economic, Social and Climatic Problems

This body of literature is concerned with trends in changing lifestyle decisions, particularly in response to economic tensions, social isolation, and feelings of climatic responsibility. A key component of this literature is feelings of dissatisfaction with the status quo for housing, with average house sizes in the US nearly tripling in size in the last half century (Lessard 2022). This increase in housing size has rendered home ownership unaffordable for many, leading them to turn towards smaller living arrangements (Kitzmann, 2023). Housing has also, through the process of becoming larger, become more isolated as well. Large single family houses on oversized lots as well as apartment buildings make it more difficult for neighbors to connect (Carbone et al. 2021). Finally, the increase in housing size has also seen a corresponding increase in environmental footprint, offsetting gains from more sustainable modern building techniques (Kitzmann, 2023). This has further motivated environmentally conscious individuals to prefer smaller housing and has prompted advocates to push for smaller housing as one of the easiest ways to reduce environmental impact (Kitzmann, 2023). While most agree these factors are increasing in importance, it is still uncertain whether they could have the mainstream appeal to create meaningful change, especially in locations such as Quebec where heating is cheap and clean and long winters makes smaller living more isolating rather than less (Lessard, 2021).

Potential of Unconventional Housing Forms

Along with the growing prevalence of smaller and traditionally “unconventional” housing forms (unconventional from a modern perspective), several questions arise as to the advantages of these housing forms and their various types. The most evident advantage is the financial gains from living smaller, effectively resulting in a lower per capita housing space and therefore housing cost (Kitzmann, 2021). Another advantage is the introduction of new regulatory frameworks that position these housing forms as officially sanctioned options rather than informal alternatives for a small minority (Cipkar 2023). These forms of housing also often attract individuals seeking a stronger community due to more community focused designs. Along with a greater sense of community, they have allowed more flexibility for seniors who would like to remain independent while still living close to family or living assistance (Carbone et al 2021). Along with these advantages, there is also the question of which type is best. Tiny houses have gained immense popularity recently, fueled by a robust media presence, but Lessard (2021) argues that their relatively high costs due to land costs and tendency to be built in rural areas limit their appeal. Garage conversions are also seen as an appealing opportunity, but this comes at the cost of parking space, which is valued by homeowners and often required by municipalities (Brown et al. 2017). Both shortcomings point to the advantage of ADUs, as they do not require additional land and often don't require parking either, especially if located near transit stations (Kim et al. 2023a).

Housing Policy Implementation and Effects

Increased adoption of ADUs has led many governments to attempt to regulate their construction. There are several debates about the ideal way to permit ADUs. The first is which scale of government should be involved. Many argue that municipalities should have the freedom to do what they want, but increasing trends in what is referred to in the US as "state preemption" challenge this belief, citing that states pay the cost of a housing crisis perpetuated by restrictive municipal legislation (Wielga 2024). There are also questions about where and how large ADUs should be, with many advocates such as ADUsearch promoting their own recommended policies (Cipkar 2022). Regardless of policy approach, it is important to measure the effectiveness of these strategies. The literature has identified three main methodologies for doing so. The first, and the one that will be adopted in this paper, is a GIS analysis of potential permittable sites (Kim D. et al. 2023a, Cipkar S. 2023). Advantages of this method are its objectiveness, universality, and scale (as many locations do not have many ADU owners to interview). Other methodologies include surveys of ADU owners and potential owners (Crane 2020, Kim et al. 2023b) and comparing differences between existing units, such as cost, size, and location (Lessard 2021).

Methodology

Overview

This study will utilize geographic information systems (GIS) to determine the permittable lots for each policy and the locational characteristics of each lot. This GIS analysis will be conducted using python with the geospatial library GeoPandas, as this will be ideal for creating easily repeatable and automatable workflows for multiple levels of each policy to be run on several municipalities. Cadaster and Overture data will be used for lot sizes and building footprints, as well as data from municipalities for other factors such as utility connections, municipal boundaries, street networks, and floodplains.

Methods

The three principal policies that will be taken into consideration in this analysis will be floor area ratio (FAR), minimum unit sizes, and number of units per lot size. These policies came from both implemented and proposed ADU legislation in Canada and the US. British Columbia relies on the number of units per lot size (British Columbia 2025), the proposed National Minimum Regulation from the project ADUsearch by the University of Windsor and Resimate (Cipkar 2022) advocates for FAR requirements, and Colorado makes use of minimum unit sizes (Colorado 2024). For instance, the number of permittable lots will be calculated for a FAR of 10%, 20%, 30% etc. which will give an indication of the impact of increasing FAR restrictions on ADU development. This kind of analysis will be run for every factor to determine the effect of increasingly stringent legislation for each.

On top of the 3 principal policies, additional characteristics that will be considered will be ADU type (detached, attached extension, or attached renovation), allowing ADUs in front yards or back yards only, parking requirements, setback requirements (how close to the street is allowed?), height limits, whether lots have utility connections, and zoning (only in single family or multi-family too). Factors that will be considered to find valid lots (unconditional on the policy tested) are that they are within municipal boundaries, are not heritage protected sites, are not within a flood plain, and have a pre-existing structure built on them (no vacant lots). These additional factors were chosen based on the studies by Kim et al. (2023a) in Southern California and Cipkar (2023) in Windsor, Ontario, which conducted similar analyses.

Additionally, lots will be evaluated based on their walkability, bikeability, transit connections, and density. Data for these indicators will come from the Walkscore project and the 2021 census by Statistics Canada. This will provide insights into the advantages and disadvantages of some policies over others beyond simply the number of permittable units. For example, the results may show that FAR restrictions limit ADUs in high density areas more than minimum unit size restrictions.

Positionality

Through my education in Urban Studies, I have come to regard data driven methods such as GIS as more valuable and I recognize the influence of my experience with these methods in choosing a project that relies on them. As a result of my training I have approached the question of how best to support ADU policy makers as a quantitative problem to be solved by broad, technical solutions, which I understand does not represent the full picture and may not be sufficient in many cases. Additionally, I recognize the impact of my identity as a cisgender white male from a western country in my acceptance of top down, professionalized solutions which I am supporting through my research. In doing so I recognize I am discrediting informal grassroots solutions through my lack of recognition of these movements.

Annotated Bibliography

Brown, A. E., Mukhija, V. & Shoup, D. (2017). Converting Garages into Housing. https://doi.org/10.1177/0739456x17741965

Carbone, J. T., Clift, J., Wyllie, T. & Smyth, A. (2021). Housing Unit Type and Perceived Social Isolation Among Senior Housing Community Residents. https://doi.org/10.1093/geront/gnab184

Cipkar, S., MacLeod, C., & Gignac-Eddy, A. (2022, August 16). *Resource centre*. ADUSearch. https://adusearch.ca/resource-centre/

Cipkar, S., Maoh, H., Dimatulac, T., Fathers, F., Arcis, S. & Smit, A. (2023). Beyond perception: Spatial analysis of detached ADU potential on residential lots in Windsor, Ontario. *The Canadian Geographer / Le Géographe canadien*. https://doi.org/10.1111/cag.12840

- In this article Cipkar explains a GIS model created to identify the amount of buildable ADUs in Windsor, Ontario
- Cipkar argues that ADUs are an important tool for diversifying the types of housing available and are valuable for increasing the supply of low income housing
- Cipkar examines various attempts to quantitatively assess ADU housing policy and potential supply, coming to the conclusion that there is not yet any large scale study in this area
- This paper was later used as a proof of concept to build ADUsearch.ca, a program that takes the methodology of this paper and applies it at a national scale
- This paper offers a useful starting point from which to orient the research proposed in this proposal. The lack of taking into consideration the impact of by-laws of various leniency in zoning standards offers an avenue upon which this paper can be expanded

Cohen, M. J. (2021). New Conceptions of Sufficient Home Size in High-Income Countries: Are We Approaching a Sustainable Consumption Transition?. *Housing Theory and Society, 38*. https://doi.org/10.1080/14036096.2020.1722218

Colorado General Assembly (2024). Retrieved from https://leg.colorado.gov/bill_files/42079/download

Crane, R. E. (2020). Is Granny in that Flat?: How Regulations Shape the Construction and Use of Accessory Dwelling Units in Los Angeles. https://escholarship.org/uc/item/2wh204vz

- This paper primarily seeks to determine why ADUs are built in some areas rather than others, as well as what ADUs are actually used for and if they contribute to the supply of affordable housing
- Crane found that high rent areas had lower quantities of ADUs, and neighborhoods where people already lived with non-relatives had more ADUs
- Crane also found that many ADU owners only considered renting to relatives or used the space themselves as an office, storage or guest room
- Crane also found that most ADU owners who rent them out use their property as a long term rather than short term rental
- This paper offers valuable insight into the effectiveness of ADU construction at combatting housing affordability issues, which will be an important caveat to note in the proposed research

Gabbe, C. J. (2019). Changing Residential Land Use Regulations to Address High Housing Prices: Evidence From Los Angeles. *Journal of the American Planning Association*, *85*(2), 152–168. https://doi.org/10.1080/01944363.2018.1559078

Kim, D., Baek, S., & Garcia, B. (2023). *Accessory Dwelling Unit (ADU) Potential in the SCAG Region*. https://scag.ca.gov/sites/default/files/2024-05/final_report_scag_v2.pdf

- This study aimed to estimate the number of ADUs that can be constructed in southern California under various policy scenarios
- The results found a significant difference in availability under different policies
- The policy changes recommended in this paper can be taken into consideration when planning the policy changes to consider in the proposed research

Kim, D., Baek, S., Garcia, B., Vo, T., Wen, F., Kim, D., Baek, S., Garcia, B., Vo, T. & Wen, F. (2023). The influence of accessory dwelling unit (ADU) policy on the contributing factors to ADU development: an assessment of the city of Los Angeles. *Journal of Housing and the Built Environment, 38*(3). https://doi.org/10.1007/s10901-022-10000-2

- This paper looked to examine the effect of a change in ordinances in Los Angeles concerning ADU development, particularly looking to see if it increased the diversity of locations in which ADUs were constructed
- The ordinance prohibited local barriers to ADU construction, allowing only by-laws that increased access to ADUs
- The results found that local intervention improved diversity of ADU adoption, particularly in areas close to transit connections

Kitzmann, R. (2023). Home swapping as instrument for more housing sufficiency!. *International Journal of Housing Policy*. https://doi.org/10.1080/19491247.2023.2269619

Lehner, M., Richter, J., Kreinin, H., Mamut, P., Vadovics, E., Henman, J., Mont, O. & Fuchs, D. (2024). Living smaller: acceptance, effects and structural factors in the EU. *Buildings & Cities*. https://doi.org/10.5334/bc.43

Lessard, G. (2018). *Accessory Dwelling Units (ADU) | Principles and Best Practices*. https://doi.org/10.13140/RG.2.2.23243.67360

- This article provides a brief summary of ADUs in Canada as well as some guidelines for cities as to how best to institute a policy on ADUs
- This brought up several points that can potentially be used as differentiating factors in the proposed research, such as presence of trees, utility connections, alley access, occupancy requirements, and the possibility of allowing ADUs first on corner lots

Lessard, G. (2022). The scaling up of the tiny house niche in Quebec – transformations and continuities in the housing regime. *Journal of Environmental Policy & Planning*, *24*(6), 625–639. https://doi.org/10.1080/1523908X.2021.2022464

- This paper examines the case study of Quebec to see how tiny home development has changed, primarily from the developer's viewpoint
- It outlines several Quebec and Canada specific concerns for smaller houses which may be useful in the proposed research, such as isolation in the long winters and the disincentive to transition to more sustainable housing due to cheap clean energy
- Lessard also discusses how most tiny homes in Quebec are built in rural areas and many are used as secondary dwellings for recreational activities such as hiking and skiing, and that when built on single family lots in rural areas these tiny houses are not much cheaper than nearby "normal" houses, which suggests an advantage of the ADU model of building in urban or suburban areas on already purchased land

Ministry of Housing and Municipal Affairs, Policy Bulletin (2025). British Columbia. Retrieved from https://www2.gov.bc.ca/assets/gov/housing-and-tenancy/tools-for-government/local-governments-and-housing/bill_25_ssmuh_policy_bulletin.pdf

Wielga, C. (2024). State preemption of land use regulations : the case of accessory dwelling units. . https://doi.org/10.32469/10355/105942